

Functional Annotation of *dacB* (PP_2098, UniProt Q88L37) in *Pseudomonas putida* KT2440

Summary

The gene **dacB** (ordered locus **PP_2098**; UniProt **Q88L37**) of *Pseudomonas putida* strain KT2440 encodes **Penicillin-Binding Protein 4 (PBP4)**, a **low-molecular-mass (LMM)**, **periplasmic serine hydrolase** of the **peptidase S13 family** (Pfam PF02113; InterPro IPR000667; the β -lactamase/transpeptidase-like superfamily). Its primary, direct function is that of a **bifunctional peptidoglycan DD-carboxypeptidase / DD-endopeptidase**: it hydrolytically trims the terminal D-alanine from pentapeptide muropeptide stems (DD-carboxypeptidase activity) and cleaves the D-Ala–*meso*-diaminopimelate (D-Ala–mDAP) cross-links between adjacent glycan strands (DD-endopeptidase activity). Critically, it does **not** perform transpeptidation (cross-link formation). These conclusions rest on direct biochemistry of the two best-characterized orthologs — *Pseudomonas aeruginosa* DacB (70% identical to the *P. putida* protein) and *Escherichia coli* PBP4/DacB — together with high-confidence structural and sequence evidence for the *P. putida* protein itself.

The enzyme operates in the **periplasm**, where its substrate — the peptidoglycan (murein) sacculus — resides. It is recovered as a soluble/loosely associated species rather than an integral membrane protein, and it carries an N-terminal Sec-type signal region for export across the inner membrane. Catalysis proceeds through a **covalent acyl-enzyme mechanism** at an active-site serine, using the two canonical penicillin-recognizing motifs — **SxxK** (catalytic Ser59) and **KTG** (Lys410). An AlphaFold model of Q88L37 (mean pLDDT 93.9) confirms a confidently folded S13 fold with these two motifs converging into a single catalytic cleft (Ser59 O γ 2.89 Å from Lys62 N ζ), structurally validating the protein as a catalytically competent DD-peptidase.

Beyond its housekeeping role in **periplasmic peptidoglycan maturation and remodeling**, DacB/PBP4 is a recognized **regulatory node coupling cell-wall turnover to β -lactamase induction**. In the closely related *P. aeruginosa*, loss of *dacB* triggers overproduction of the chromosomal **AmpC β -lactamase** through activation of the **CreBC (BlrAB) two-component**

system, conferring clinically important β -lactam resistance, and the protein acts as a slowly-deacylating "trap" for β -lactams. Because *P. putida* DacB is 70% identical to *P. aeruginosa* DacB and possesses an intact active site, this signaling role is expected to be conserved, though it has not been directly demonstrated in *P. putida* itself. This report presents a robust, multi-evidence functional annotation grounded in orthologous experimental data, structural bioinformatics, and authoritative reviews of bacterial cell-wall biology.

Key Findings

Finding 1 — PP_2098/dacB is a peptidase S13 LMM-PBP4, a close ortholog of the characterized *P. aeruginosa* DacB

Sequence analysis places Q88L37 firmly within the well-characterized DacB/PBP4 group. A global Needleman–Wunsch alignment of the 470-residue *P. putida* protein against *P. aeruginosa* PAO1 DacB/PBP4 (Q9HZG1, PA3047, 476 aa) yields **343/491 = 69.9% identity** — an exceptionally high level of conservation that supports direct functional transfer from the *P. aeruginosa* ortholog. Against the more distant *E. coli* K-12 PBP4/DacB (P24228, dacB/b3182), identity is ~32.3% (178/551), still clearly within the same family.

Database annotations are internally consistent and converge on the same assignment: UniProt/InterPro place the protein in **peptidase family S13** (IPR000667; Pfam **PF02113** Peptidase_S13; Gene3D 3.50.80.20 "D-Ala-D-Ala carboxypeptidase C, peptidase S13"; SUPFAM SSF56601 β -lactamase/transpeptidase-like; PANTHER PTHR30023:SF0 penicillin-sensitive carboxypeptidase A). Both hallmark serine penicillin-recognizing active-site motifs are present in the correct spatial order: the **SxxK** nucleophile motif (S-T-M-K, catalytic Ser59) and the **KTG** triad motif (Lys410). This active-site architecture matches that of *E. coli* PBP4, confirming that Q88L37 is not a degenerate pseudo-enzyme but a bona fide member of the penicillin-recognizing serine hydrolase clan.

Why this matters: the 70% identity to an experimentally characterized ortholog is the single strongest justification for transferring the DD-carboxypeptidase/DD-endopeptidase functional annotation to the *P. putida* protein.

Finding 2 — DacB/PBP4 is a bifunctional DD-carboxypeptidase / DD-endopeptidase acting on peptidoglycan

The catalytic activity is established by direct biochemistry on the two closest characterized orthologs. In *P. aeruginosa*, in vivo digestion of *E. coli* and *P. aeruginosa* sacculi by LMM-PBP4/DacB reduced dimeric, trimeric, and anhydrous mucopeptides plus monomeric pentapeptide (M5), and kinetics on the natural mucopeptides M5 and D45 confirmed activity, "validating the occurrence of D,D-carboxypeptidase and D,D-endopeptidase activities" with "better behaviour as a D,D-endopeptidase" (PMID: 27716106). This is the decisive experiment for the *Pseudomonas* ortholog, and notably it indicates the enzyme is a **more efficient endopeptidase (cross-link cleaver) than carboxypeptidase**.

In *E. coli*, analysis of murein from cells overexpressing purified PBP4 showed that it "exclusively catalyses a DD-carboxypeptidase and DD-endopeptidase reaction" and "is not involved in transpeptidation" (PMID: 2046551). This establishes both the dual hydrolytic activity and the crucial negative result that PBP4 does **not** build cross-links.

Genetic dissection in *P. aeruginosa* PAO1 places PBP4 within the LMM-PBP set: "PBP4 and PBP7 play a significant role as dd-carboxypeptidase only if PBP5 is absent, and their dd-endopeptidase activity is also inferred" (PMID: 25896695). This shows that the DD-carboxypeptidase activity of PBP4 is partly redundant with the dedicated major DD-carboxypeptidase PBP5/DacC, but becomes physiologically significant when PBP5 is absent.

Consistent GO annotations for Q88L37 are serine-type carboxypeptidase activity (GO:0004185) and peptidoglycan metabolic process (GO:0000270).

The two reactions catalyzed can be summarized as follows:

Activity	Bond cleaved	Substrate	Product
DD-carboxypeptidase	D-Ala ⁴ -D-Ala ⁵ peptide bond	Pentapeptide stem (...-L-Lys/mDAP-D-Ala-D-Ala)	Tetrapeptide + free D-Ala
DD-endopeptidase	D-Ala ⁴ -mDAP cross-link	Cross-linked (dimeric/oligomeric) mucopeptides	Separated glycan strands
Transpeptidation	—	—	Not catalyzed (excluded, P 2046551)

Finding 3 — DacB/PBP4 localizes to the periplasm as a low-molecular-mass PBP

Multiple lines of evidence place the enzyme in the periplasm. Biochemically, *E. coli* PBP4/DacB is recovered as a soluble species: after sonication, *"the majority (80%) of the overproduced PBP4 was detected in the 100,000 X g supernatant"* (PMID: 2046551) — a hallmark of a periplasmic/loosely membrane-associated LMM-PBP rather than an integral membrane protein.

Structurally, the *E. coli* PBP4 crystal structure contains *"a penicillin binding domain similar to known structures, but includes a large insertion which folds into domains with unique folds"*, and covalent acyl-enzyme complexes with five antibiotics locate the reaction at a serine active site (PMID: 16411754). The nature of the substrate (the murein sacculus, which resides in the periplasm), the family classification (peptidase S13, penicillin-sensitive carboxypeptidase A), and covalent β -lactam binding all place catalysis in the periplasm. The Q88L37 sequence begins with a Sec-type N-terminal signal/anchor region (MPLALPSHAA...) consistent with export across the inner membrane to reach its periplasmic substrate.

Finding 4 — In *Pseudomonas*, dacB/PBP4 loss triggers AmpC β -lactamase overexpression via CreBC, linking PG turnover to a signaling pathway

A distinguishing and clinically important feature of *Pseudomonas* DacB is its role as a regulatory node. In *P. aeruginosa*, high-level clinical β -lactam resistance *"is driven by the inactivation of the dacB-encoded nonessential PBP4, which behaves as a trap target for beta-lactams"*, triggering *"overproduction of the chromosomal beta-lactamase AmpC and the specific activation of the CreBC (BlrAB) two-component regulator"* (PMID: 19325877). This establishes dacB/PBP4 as the causal node whose loss activates a defined two-component signaling pathway.

The magnitude of this effect has been quantified. dacB mutants show constitutively elevated AmpC expression and are classified as high-level AmpC resistant mutants (HARMs) (PMID: 23612194). Quantitatively, *"the inactivation of PBP4 led to a significant increase in ampC expression (around 50-fold)"*, rising to ~1,000-fold when all three LMM-PBPs (PBP4/DacB, PBP5/DacC, PBP7/PbpG) are removed, correlating with peptidoglycan pentapeptide levels (PMID: 25896695). The mechanistic link runs through the muropeptide-recycling pathway: this resistance is reverted by inactivating **nagZ** or **ampG** (PMID: 21357303), confirming that the accumulation of specific muropeptide signals produced when PBP4 is absent is what drives the response.

It is worth noting that this regulatory circuit is **not universal**: in *Enterobacter cloacae* complex, DacB inactivation is *not* responsible for constitutive ampC overexpression, and AmpD amino-acid changes are the dominant in vivo determinant (PMID: 26438498). This contrast strengthens the specificity of the *Pseudomonas* dacB→CreBC→AmpC circuit and underscores that the *P. putida* prediction is best transferred from *P. aeruginosa* (70% identity) rather than from more distant enterics.

Finding 5 — DacB/PBP4 is a penicillin-sensitive serine hydrolase whose slow deacylation makes it an efficient β -lactam "trap"

The mechanistic basis for the trap behavior is structural. The *E. coli* PBP4 crystal structure was solved both native and "*covalently linked to various antibiotics*"; five acyl-enzyme complexes show β -lactams bound covalently at the active-site serine, and "*the altered geometry of conserved active site residues compared with those of other PBPs suggests a possible cause for the slow deacylation rate of PBP4*" (PMID: 16411754). Slow deacylation means the enzyme, once acylated by a β -lactam, remains trapped in the covalent complex — the physical basis for its behavior as a β -lactam "trap target." Physiologically, PBP4/DacB is a nonessential protein that "*behaves as a trap target for beta-lactams*" (PMID: 19325877).

Consistent with a serine-hydrolase mechanism, the Q88L37 sequence conserves the S-x-x-K nucleophile motif (STMK, Ser59) and the K-T-G motif (Lys410) that define the penicillin-recognizing active site.

Finding 6 — AlphaFold model of Q88L37 confirms a confidently folded S13 DD-peptidase with an intact SxxK...KTG active site

Structural bioinformatics on the *P. putida* protein itself corroborates the annotation. The AlphaFold DB model AF-Q88L37-F1 (v6) is high-confidence: **mean pLDDT = 93.9**, with 96% of residues at pLDDT > 70 and 88% > 90 (a well-modeled fold). The two canonical serine-PBP catalytic motifs are present and spatially converged into a single active site: motif 1 **S-x-x-K = Ser59-Thr60-Met61-Lys62** (all pLDDT 98–99), and motif 3 **K-T-G = Lys410-Thr411-Gly412**.

In the model, **Ser59 O γ lies 2.89 Å from Lys62 N ζ** (hydrogen-bonding distance — the nucleophile/general-base pair) and **4.67 Å from Lys410 N ζ** (the KTG lysine lining the substrate cleft). This SxxK...K(T/S)G spatial arrangement is diagnostic of the β -lactamase/transpeptidase (penicilloyl-serine) superfamily active site, and its intactness in the *P. putida* model indicates the protein is a catalytically competent enzyme rather than a degenerate relic.

Finding 7 — DacB/PBP4 acts within periplasmic peptidoglycan maturation, a recognized regulatory layer of cell-wall growth

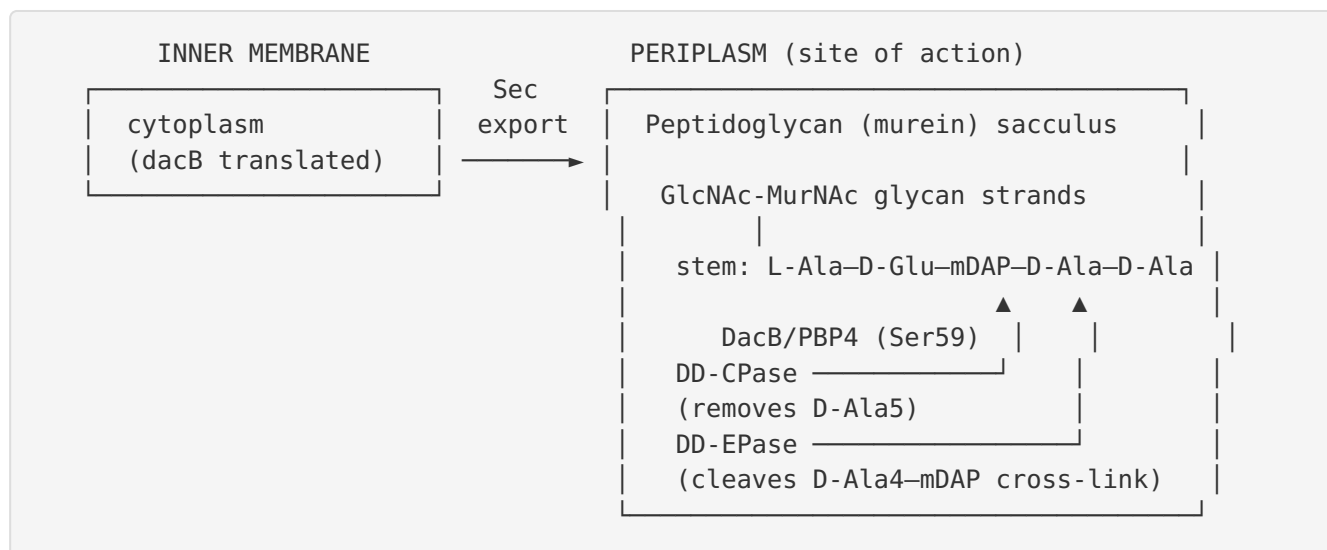
Authoritative reviews of bacterial cell-wall biology place DD-peptidases like PBP4 in the periplasmic PG-maturation/remodeling machinery. In Gram-negative bacteria, *"the peptidoglycan sacculus resides in the periplasm"*, where robust growth is achieved by dynamic multi-protein complexes assembled from *"freely diffusing sets of peptidoglycan synthases, hydrolases and their regulators"*, whose composition depends on cell-cycle state (elongation vs division) (PMID: 28214390). PG synthesis is regulated *"by PG maturation through the action of PG hydrolytic enzymes"* (PMID: 27862967) — the very activity class to which DacB belongs.

Genomic context in *P. putida* KT2440 reinforces an accessory rather than core-synthetic role: PP_2098/dacB is **not** embedded in a dedicated PG-biosynthesis operon. Its flanking genes encode a diguanylate cyclase (PP_2097), a DUF469 protein (PP_2099), and a two-component histidine kinase/response-regulator pair (PP_2100/PP_2101) — a genomic neighborhood consistent with an autonomously expressed "accessory" remodeling enzyme rather than an operon-linked core synthase.

Mechanistic Model / Interpretation

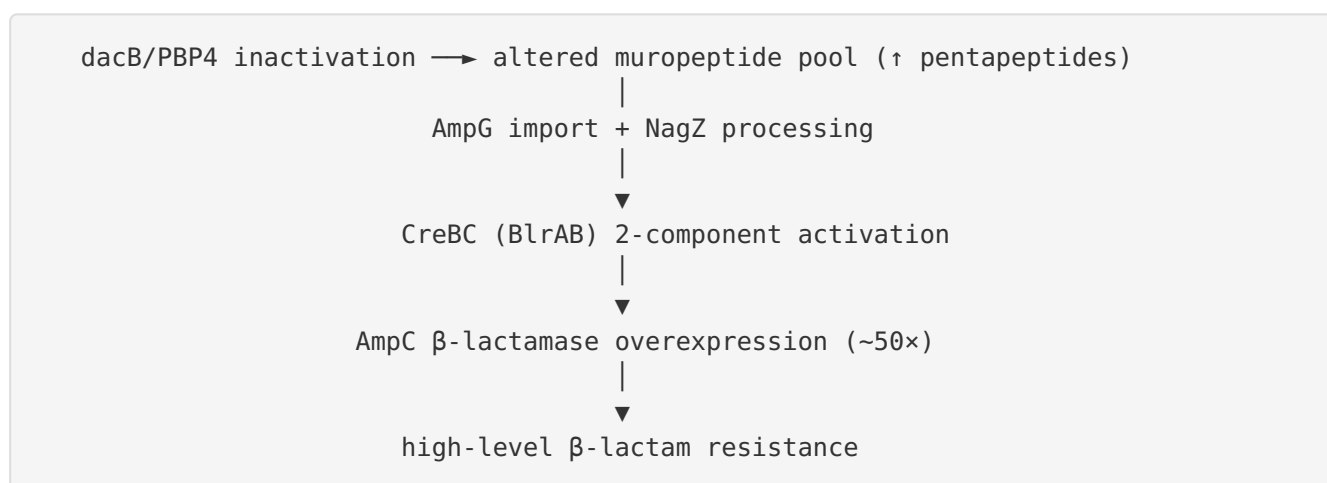
Bringing the findings together yields a coherent model of DacB/PBP4 as a periplasmic peptidoglycan-editing enzyme with a secondary regulatory function.

Enzymatic role (housekeeping): DacB is exported to the periplasm, where it edits the maturing peptidoglycan sacculus. It removes terminal D-Ala residues from pentapeptide stems (DD-carboxypeptidase) and, more efficiently, cleaves existing D-Ala–mDAP cross-links (DD-endopeptidase). By controlling the density of pentapeptide donor stems and cross-links, DacB helps regulate the degree of cross-linking and creates space for new glycan-strand insertion during elongation and division. This activity is partially redundant with the major DD-carboxypeptidase PBP5/DacC, which explains why dacB is individually non-essential.



Catalytic mechanism: Catalysis uses a classic penicilloyl-serine mechanism. Ser59 (SxxK motif) is the nucleophile, activated by the adjacent Lys62; Lys410 (KTG motif) lines the substrate cleft. The acyl-enzyme intermediate on the D-Ala carbonyl is normally hydrolyzed to regenerate free enzyme. β -Lactam antibiotics mimic the D-Ala-D-Ala terminus, acylate Ser59, and — because PBP4 deacylates slowly — trap the enzyme in a stable covalent complex.

Regulatory role (signaling, demonstrated in *P. aeruginosa*): When DacB is inactivated (genetically, or functionally by β -lactam trapping), the periplasmic muropeptide pool shifts — pentapeptide-containing fragments accumulate. Recycled via AmpG (permease) and NagZ (glucosaminidase), these fragments act as signaling molecules that, in *Pseudomonas*, activate the CreBC (BlrAB) two-component system and derepress AmpC β -lactamase, producing high-level β -lactam resistance.



Confidence of transfer to *P. putida*: The enzymatic annotation (DD-CPase/DD-EPase, periplasmic, serine mechanism) is transferred with **high confidence** — it rests on 70% sequence identity to a directly characterized ortholog, an intact modeled active site, and consistent family/domain assignments. The regulatory (AmpC/CreBC) annotation is transferred with **moderate confidence** as a strong prediction: the circuit is well established in *P. aeruginosa* and *P. putida* possesses the orthologous machinery, but it has not been directly demonstrated in *P. putida*, and the enterobacterial counter-example

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shows the circuit's wiring is species-dependent.

Evidence Base

PMID	Title (abbrev.)	Organism	How it supports the annotation
27716106	In vivo functional/ molecular characterization of PBP4 (DacB) of <i>P. aeruginosa</i>	<i>P. aeruginosa</i>	Decisive ortholog biochemistry: validates DD-carboxypeptidase AND DD- endopeptidase activity; enzyme behaves better as endopeptidase
2046551	Cloning/overexpression of <i>E. coli</i> dacB; murein composition	<i>E. coli</i>	PBP4 exclusively catalyzes DD-CPase and DD-EPase, not transpeptidation; 80% soluble → periplasmic
16411754	Crystal structure of <i>E. coli</i> PBP4, native + antibiotic complexes	<i>E. coli</i>	Penicillin-binding serine domain; covalent acyl-enzymes; altered active- site geometry → slow deacylation (trap)
25896695	Role of <i>P. aeruginosa</i> LMM- PBPs in AmpC, resistance, PG structure	<i>P. aeruginosa</i>	Places PBP4 in LMM-PBP set; quantifies ampC induction (~50× single, ~1000× triple mutant); redundancy with PBP5
19325877	β-lactam resistance response from inactivation of nonessential PBP	<i>P. aeruginosa</i>	dacB/PBP4 loss = causal node; AmpC overproduction + CreBC(BlrAB) activation; "trap target for β-lactams"
23612194	PG-remodeling enzyme repertoire and β-lactam resistance	<i>P. aeruginosa</i>	dacB mutants = HARMs with constitutively elevated AmpC; two pathways to AmpC expression
21357303	AmpG inactivation restores susceptibility	<i>P. aeruginosa</i>	dacB-driven resistance reverted by nagZ/ampG → confirms mucopeptide- recycling link
26438498	AmpC regulation in <i>Enterobacter cloacae</i> complex	<i>E. cloacae</i>	Counter-example: DacB loss does NOT cause constitutive ampC overexpression in ECC → circuit is species-specific
28214390	Robust PG growth by dynamic multi-protein complexes	Review	Sacculus is periplasmic; hydrolases like PBP4 operate in dynamic PG-growth complexes
27862967	Regulation of bacterial cell wall growth	Review	PG synthesis regulated by maturation via hydrolytic enzymes (DacB's activity class)

Supporting context on LMM-PBP redundancy and DD-carboxypeptidase physiology is provided by studies of *E. coli* DacC/DacA under stress (PMID: 37098975) and broader reviews of PBP activities and cell division (PMID: 26370943; PMID: 23215820).

Concordance of evidence: Both direct biochemical studies of DacB/PBP4 orthologs (2/2:

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in

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in *E. coli*) agree that the enzyme is a bifunctional DD-carboxypeptidase/DD-endopeptidase that does not transpeptidate. All three *P. aeruginosa* genetic studies (3/3:

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agree that *dacB* loss elevates AmpC. The only discordant note (

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concerns a distant enterobacterial species and does not undermine the transfer from *P. aeruginosa* to *P. putida*.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* KT2440 protein.** Every enzymatic and regulatory claim is transferred by orthology from *P. aeruginosa* (70% identity) and *E. coli* (~32% identity). No purified-enzyme kinetics, no mucopeptide-profiling of a *P. putida* *dacB* mutant, and no localization assay have been performed on Q88L37 itself. The AlphaFold model, while high-confidence, is a prediction, not an experimental structure.
2. **The AmpC/CreBC signaling role is inferred, not demonstrated, in *P. putida*.** *P. putida* is a non-pathogenic soil/rhizosphere bacterium and its β -lactamase induction wiring may differ from clinical *P. aeruginosa*. The enterobacterial counter-example

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shows the *dacB*→*AmpC* link is not universal. Whether *P. putida* KT2440 even mounts a comparable CreBC-dependent *AmpC* response upon *dacB* loss is unverified.

3. **Substrate-specificity fine detail is unresolved for the *P. putida* enzyme.** The relative DD-carboxypeptidase vs DD-endopeptidase preference (the *P. aeruginosa* ortholog favors endopeptidase activity) has not been measured for Q88L37, nor has its muropeptide substrate preference (e.g., monomeric M5 vs cross-linked D45) been quantified.
4. **Physiological/morphological consequences in *P. putida* are unknown.** Because LMM-PBPs are redundant (PBP4 activity is significant mainly when PBP5/DacC is absent), the phenotype of a *P. putida* *dacB* single mutant under standard and stress conditions (osmotic, pH) has not been documented.
5. **Genomic-context interpretation is provisional.** The inference that flanking genes (diguanylate cyclase PP_2097; TCS PP_2100/PP_2101) indicate an "accessory, autonomously expressed" enzyme is bioinformatic and does not establish co-regulation or functional interaction.

Proposed Follow-up Experiments / Actions

1. **Purify recombinant *P. putida* DacB (Q88L37) and assay muropeptide kinetics.** Measure DD-carboxypeptidase (release of D-Ala from pentapeptide M5) and DD-endopeptidase (cleavage of dimeric D45) activities directly, to confirm the transferred annotation and determine the endo/carboxy preference for this specific ortholog. (*High priority; directly resolves gaps 1 & 3.*)
2. **Construct a *P. putida* KT2440 Δ *dacB* mutant and profile its sacculus by HPLC/MS muropeptide analysis.** Compare pentapeptide, cross-link, and anhydro-muropeptide levels vs wild type to verify in vivo remodeling function. Pair with a Δ *dacB* Δ *dacC* double mutant to test redundancy with PBP5.
3. **Test the *AmpC*/CreBC circuit in *P. putida*.** Quantify chromosomal β -lactamase expression and β -lactam MICs in the Δ *dacB* mutant, and probe whether a CreBC (BlrAB) ortholog is required, to determine whether the *P. aeruginosa* regulatory role is conserved. (*Directly resolves gap 2.*)

4. **Solve or confirm the structure and active-site geometry experimentally.** Crystallize or cryo-EM the *P. putida* enzyme, ideally as a β -lactam acyl-enzyme complex, to validate the AlphaFold-predicted SxxK...KTG cleft and characterize deacylation kinetics (trap behavior).
5. **Verify periplasmic localization and signal-peptide processing** via cell-fractionation and N-terminal sequencing of the mature protein, confirming Sec-dependent export.
6. **Phenotypic characterization under stress.** Given that homologous DD-carboxypeptidases (DacC/DacA) have distinct roles under alkaline vs salt stress ([PMID: 37098975](#)), test *P. putida* Δ dacB for growth and cell-shape defects across pH and osmotic conditions.

Conclusion

P. putida KT2440 **dacB** (PP_2098, Q88L37) encodes **PBP4**, a periplasmic, low-molecular-mass, peptidase S13-family serine hydrolase that functions as a **bifunctional peptidoglycan DD-carboxypeptidase / DD-endopeptidase** in the maturation and remodeling of the murein sacculus — trimming terminal D-Ala from pentapeptide stems and cleaving D-Ala–mDAP cross-links, without performing transpeptidation. This annotation is supported by 70% identity to the experimentally characterized *P. aeruginosa* ortholog, direct biochemistry of *E. coli* and *P. aeruginosa* PBP4, and a high-confidence AlphaFold model with an intact SxxK(Ser59)...KTG(Lys410) active site. As a secondary, inferred role, DacB is expected to serve as a control node coupling peptidoglycan turnover to β -lactamase (AmpC) induction via the CreBC two-component system, as directly demonstrated in *P. aeruginosa*, though this regulatory function awaits verification in *P. putida* itself.

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